

Exact Conservative-Field Projection Constants

Cycles, cacti, complete graphs, and complete bipartite graphs
for arXiv:2309.09313

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Status. Candidate partial result, likely valid. The unrestricted finite-graph question remains open, but the optimal constant is computed exactly for weighted cycles, weighted cacti, unweighted complete graphs, unweighted complete bipartite graphs, and every graph assembled from these pieces at cut vertices.

1 Source question and normalization

Schlumprecht asks for the smallest norm of an extension of the integral operator from conservative vector fields on a finite weighted graph to all vector fields [1]. The exact source passage is reproduced below.

Question: How well is $\text{CVF}(G)$ complemented in $\text{VF}(G)$ Equivalently, what is the smallest constant $C \geq 1$, so that the operator

$$I : \text{CVF}(G) \rightarrow \text{Lip}_0(V(G), d)$$

can be extended to an operator

$$\tilde{I} : \text{VF}(G) \rightarrow \text{Lip}_0(V(G), d).$$

Figure 1: The question on page 33 of arXiv:2309.09313v1.

Fix one orientation of each edge. Write $\text{VF}(G) = \ell_\infty(E(G))$, and identify $\text{CVF}(G)$ with the subspace of fields whose weighted integral around every cycle is zero. Define the relative projection constant

$$\lambda(G) = \inf\{\|P\|_\infty : P^2 = P, \text{ ran } P = \text{CVF}(G)\}.$$

This is exactly the constant in the source question. Indeed, if $\tilde{I} : \text{VF}(G) \rightarrow \text{Lip}_0(V(G))$ extends integration, then $P = \nabla \tilde{I}$ is a projection onto $\text{CVF}(G)$ with the same norm. Conversely, IP is an extension whenever P is such a projection. Both norm equalities follow because $\nabla : \text{Lip}_0(V(G)) \rightarrow \text{CVF}(G)$ and its inverse I are isometries.

2 Results

For a weighted cycle C , let its positive edge lengths be $a_1, \dots, a_r$, and put $A = \sum_i a_i$. Since the edge lengths come from a geodesic graph metric, $a_i \leq A/2$.

Theorem 1 (Weighted cycle). *If some $a_i = A/2$, then $\lambda(C) = 1$. Otherwise*

$$\lambda(C) = 1 + \left(\sum_{i=1}^r \frac{a_i}{A - 2a_i} \right)^{-1}. \tag{1}$$

In particular, for the unweighted cycle C_r ,

$$\lambda(C_r) = 2 - \frac{2}{r}.$$

A cactus is a connected graph in which every edge lies in at most one simple cycle.

Corollary 2 (Weighted cactus). *For a weighted cactus G ,*

$$\lambda(G) = \max(\{1\} \cup \{\lambda(C) : C \text{ is a simple cycle of } G\}),$$

where each cycle constant is given by Theorem 1. Thus trees have constant one.

Theorem 3 (Two overlapping-cycle families). *For the unweighted complete graph and complete bipartite graph,*

$$\lambda(K_n) = 2 - \frac{2}{n}, \tag{2}$$

$$\lambda(K_{p,q}) = 3 - \frac{2}{p} - \frac{2}{q} + \frac{2}{pq}. \tag{3}$$

Formula (3) also gives one when $p = 1$ or $q = 1$, as it should for a tree.

Recall that the blocks of a connected graph are its maximal 2-connected subgraphs together with its bridges. Each edge belongs to one block, and each cycle is contained in one block.

Theorem 4 (Cut-vertex gluing). *For every connected weighted graph,*

$$\lambda(G) = \max_{B \text{ a block of } G} \lambda(B). \tag{4}$$

Consequently, the preceding formulas give the exact answer for any graph whose non-bridge blocks are weighted cycles, unweighted complete graphs, or unweighted complete bipartite graphs.

Thus the result includes weighted cacti, clique trees, biclique trees, and arbitrary cut-vertex mixtures of these classes. The complete bipartite case is important because its cycle constraints overlap heavily; the answer is not merely a consequence of separating edge-disjoint cycles.

3 Proof intuition

There are three mechanisms. A weighted cycle has only one constraint, so its conservative fields form a hyperplane in ℓ_∞^r . Every projection onto a hyperplane has a rank-one defect, reducing the full operator optimization to distributing one unit of mass among the edges. The cheapest distribution gives (1).

For a highly symmetric graph, average an arbitrary projection over its automorphism group. On K_n , symmetry forces the averaged map to be the divergence projection. On $K_{p,q}$, it forces the familiar two-way additive projection: row mean plus column mean minus grand mean. Since averaging cannot increase the norm, explicit row sums give both upper and lower bounds.

Finally, blocks have disjoint edge coordinates and every cycle stays inside a single block. Conservativity therefore becomes an ℓ_∞ -direct sum condition, for which the optimal constant is the largest block constant.

4 The weighted-cycle computation

Orient the cycle consistently. Its conservative fields are

$$H = \ker \varphi, \quad \varphi(x) = \sum_{i=1}^r a_i x_i.$$

Every projection from ℓ_∞^r onto H has the form

$$P = I - u \otimes \varphi, \quad \varphi(u) = 1.$$

Indeed, $I - P$ vanishes on the codimension-one space $\ker \varphi$, so it factors through φ ; idempotence fixes the normalization.

Put $t_i = a_i u_i$ and $c_i = (A - a_i)/a_i$. The i -th row sum of P is

$$R_i = |1 - t_i| + c_i |t_i|, \quad \sum_i t_i = 1. \quad (5)$$

This argument works over both the real and complex scalar fields. If $a_i < A/2$, then $c_i > 1$, and the triangle inequality gives

$$R_i \geq 1 + (c_i - 1)|t_i|.$$

Consequently, if all $R_i \leq L$, then

$$1 = \left| \sum_i t_i \right| \leq \sum_i |t_i| \leq (L - 1) \sum_i \frac{1}{c_i - 1} = (L - 1) \sum_i \frac{a_i}{A - 2a_i}.$$

This proves the lower bound in (1). Equality is attained by the positive choice

$$t_i = \frac{a_i/(A - 2a_i)}{\sum_j a_j/(A - 2a_j)}, \quad u_i = \frac{t_i}{a_i},$$

because then every row sum in (5) has the claimed value. If $a_k = A/2$, take $t_k = 1$ and all other $t_i = 0$. Every row sum is one, proving the endpoint. The unweighted specialization follows by substituting $a_i = 1$ and $A = r$.

5 Symmetry computations

5.1 Complete graphs

Represent an edge field on K_n antisymmetrically: $f_{ij} = -f_{ji}$. Put

$$d_i = \sum_k f_{ik}.$$

The canonical projection onto gradients is

$$(Pf)_{ij} = \frac{d_i - d_j}{n}. \quad (6)$$

It is a projection because for a gradient $f_{ij} = x_j - x_i$, one has $d_i = n(\bar{x} - x_i)$, and (6) returns f .

For a fixed output edge (i, j) , the input edge (i, j) has coefficient $2/n$, and the $2(n - 2)$ other edges incident with i or j have coefficient of absolute value $1/n$. Hence

$$\|P\|_\infty = \frac{2 + 2(n - 2)}{n} = 2 - \frac{2}{n}.$$

For optimality, average any projection over the vertex-permutation group S_n . The average has no larger norm. Compose it with the isometric inverse gradient to obtain an equivariant map T from edge fields to mean-zero vertex potentials. The stabilizer of a vertex i forces $T_i(f)$ to have one common coefficient on all outgoing incident edges. Edges not incident with i have coefficient zero: swapping their two endpoints fixes i and reverses their sign. Thus $T_i(f) = c \sum_k f_{ik} = cd_i$. The identity on gradients forces $c = -1/n$, so the averaged projection is uniquely (6). Its norm is therefore a lower bound for every projection, proving (2).

5.2 Complete bipartite graphs

Write a field on $K_{p,q}$ as a p -by- q matrix $x = (x_{ij})$, orienting every edge from the first part to the second. Gradients are exactly the additive matrices

$$\mathcal{A} = \{x : x_{ij} = a_i + b_j\}.$$

The two-way additive projection is

$$(Qx)_{ij} = \frac{1}{q} \sum_{\ell} x_{i\ell} + \frac{1}{p} \sum_k x_{kj} - \frac{1}{pq} \sum_{k,\ell} x_{k\ell}. \quad (7)$$

It fixes $\mathcal{A}$. The absolute row sum of (7) is

$$\begin{aligned} \frac{p+q-1}{pq} + (q-1)\frac{p-1}{pq} + (p-1)\frac{q-1}{pq} + (p-1)(q-1)\frac{1}{pq} \\ = 3 - \frac{2}{p} - \frac{2}{q} + \frac{2}{pq}. \end{aligned}$$

It remains to prove minimality. Average any projection over $S_p \times S_q$. Let $\mathcal{Z}$ be the matrices with every row sum and every column sum zero. Then the whole matrix space is $\mathcal{A} \oplus \mathcal{Z}$, and (7) is the associated projection. If R is any invariant projection onto $\mathcal{A}$ and $z \in \mathcal{Z}$, invariance under the stabilizer of an entry (i, j) shows that $(Rz)_{ij}$ is a linear combination of four orbit sums: the entry itself, the rest of its row, the rest of its column, and all remaining entries. On $\mathcal{Z}$, these four sums are respectively z_{ij} , $-z_{ij}$, $-z_{ij}$, z_{ij} . Transitivity therefore gives $Rz = cz$ for a scalar c . But $Rz \in \mathcal{A}$, while $z \in \mathcal{Z}$ and $\mathcal{A} \cap \mathcal{Z} = \{0\}$; hence $c = 0$. Thus the invariant projection is unique and equals Q . Averaging proves its minimality and establishes (3).

6 Block gluing

The edge sets of the blocks partition $E(G)$, so

$$\text{VF}(G) = \bigoplus_{B}^{\ell_{\infty}} \text{VF}(B).$$

Every cycle of a graph is contained in one block. Therefore a field is conservative on G exactly when its restriction to every block is conservative, and

$$\text{CVF}(G) = \bigoplus_{B}^{\ell_{\infty}} \text{CVF}(B).$$

Choosing optimal block projections gives $\lambda(G) \leq \max_B \lambda(B)$. Conversely, if P is any global projection, inject fields supported on one block B , apply P , and restrict the output back to B . The resulting map is a projection onto $\text{CVF}(B)$ of norm at most $\|P\|$. Taking the infimum over P and then the maximum over blocks proves (4).

7 Verification, novelty, and remaining obstruction

The companion script `code/verify_projection_constants.py` performed exact rational checks of idempotence, range constraints, and row norms on 36 weighted cycles, K_n for $2 \leq n \leq 8$, seven complete-bipartite cases, and mixed block sums. Independently formulated linear programs recovered the claimed optima in all 36 cycle cases, for K_n with $2 \leq n \leq 5$, and for six small bicliques. The script reported 190 exact assertions and 46 independent LP optimum assertions.

The general link between invariant projections, graph cycle/cut spaces, and transportation cost spaces is established in Dilworth–Kutzarova–Ostrovskii [2, 3]. Their work also warns that invariant projections need not be unique in less rigid graph families. Bohnenblust’s classical codimension-one bound and its equality theory are discussed by Kobos [4]; the weighted-cycle formula here is therefore best viewed as an explicit hyperplane calculation, not as a claim of a new general projection theorem. Rieffel gives a broader extension/projection correspondence [5].

Bounded exact-phrases, title/id, cut-space, weighted-hyperplane, complete, and complete-bipartite searches through 11 August 2026 did not locate the full collection of formulas and block synthesis stated here. Some network literature does record the norm $2(n-1)/n$ of the standard cut projection for uniform rings and complete graphs. Accordingly, this packet claims a verified exact answer for the listed families, but makes no priority claim.

For a general 2-connected graph, several independent cycle equations overlap, and the conservative subspace is no longer a single hyperplane or a disjoint coordinate sum. Symmetry averaging still produces a minimal invariant projection, but that projection can lie in a nontrivial family; uniqueness arguments used for K_n and $K_{p,q}$ then fail. Resolving that optimization uniformly is the remaining obstruction to the source’s full question.

References

- [1] T. Schlumprecht, *Transportation cost spaces and their embeddings into L_1 , a survey*, arXiv:2309.09313 (2023).
- [2] S. J. Dilworth, D. Kutzarova, and M. I. Ostrovskii, *Cycle spaces: invariant projections and applications to transportation cost*, arXiv:2305.12582 (2023).
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- [4] T. Kobos, *Hyperplanes of finite-dimensional normed spaces with the maximal relative projection constant*, Bull. Aust. Math. Soc. 91 (2015), 447–463; arXiv:1411.6214.
- [5] M. A. Rieffel, *Lipschitz extension constants equal projection constants*, Contemp. Math. 414 (2006), 147–162; arXiv:math/0508097.